Spatial Memory of Auditory Sources in Language Comprehension

Introduction. Research on sign languages (Emmorey, 1996), co-speech gestures (So et al., 2009), and eye movements during spoken-language comprehension (Staudte & Altmann, 2017) demonstrates that language users routinely draw on spatial cues in the visual environment during reference processing. These findings raise an important theoretical question: Is the reliance on spatial information restricted to referents that are visually present, or at least were visually available, or can purely auditory spatial cues also support reference resolution? Because spoken language is primarily auditory and referents are often not visually accessible, it is essential to determine whether listeners encode and retain spatial properties of auditory referents. The present study examines whether spatial representations derived solely from acoustic cues are maintained and reactivated during reference processing. We conducted two experiments in English to evaluate whether referent activation is more strongly driven by repeated names or by referent properties. **Methods.** Participants completed the task using a novel 16-loudspeaker surround-sound apparatus (Figure 1), with 10 speakers used to present auditory stimuli. In Experiment 1, participants heard two brief introductions presented sequentially from random spatial locations one from the right and the other from the left. For instance, one introduction (e.g., *"Hi, I'm John and I'm a teacher at Model Elementary School."*) could be played from one of the loudspeakers on the left, followed by a second introduction (e.g., *"Hi, my name is Isaac and I'm a student at the University of Southern California."*) played from a loudspeaker on the right. Introductions were always gender-matched within a trial. The choice of speakers in each direction was random and the order of directions was also randomized but was balanced such that half the trials for each participant were right-first and the other left-first. After the introductions, participants heard a single probe name referring to either the first or second introduction. The probe was presented either from the same spatial location as its corresponding introduction (*Direction Match*) or from the opposite location (*Direction Mismatch*). A comprehension question then appeared on a screen, and participants indicated whether the probe name matched the question (*True*) or did not (*False*). Reaction time and accuracy were recorded. A schematic of the procedure is shown in Figure 2. **Experiment 1.** Experiment 1 used a fully within-subjects 2 × 2 × 2 design manipulating Direction Match (Match vs. Mismatch), Probe Referent Order (First vs. Second), and True Probe (True vs. False). Fifty-nine undergraduate participants (44F, 15M; M_{age} = 18.98, *SD* = 0.97) were recruited through the University of South Carolina SONA pool. Data were analyzed in R 4.5.0 using mixed-effects models (R Core Team, 2023). Participants were significantly less accurate and slower on Direction Mismatch trials than on Match trials. They were also faster and more accurate when the probe indexed the second introduction. A significant interaction between Probe Referent Order and True Probe was also observed (Figure 3). These results indicate that listeners automatically maintain and activate spatial information associated with an auditory referent. They further reveal a robust recency advantage for the second referent and suggest possible inhibitory processes when retrieving information about the first referent. **Experiment 2.** Experiment 2 used the same 2 × 2 × 2 design and analytic approach but reversed the order of the probe and question. This manipulation allowed us to compare referential activation driven by repeated names (Experiment 1) versus referent properties (Experiment 2). Seventy-one participants (53F, 11M, 1NB; M_{age} = 18.46, *SD* = 1.05) were recruited. RT results replicated Experiment 1, showing direction match and recency effects. Unlike Experiment 1, neither of these effects extended to accuracy. As in Experiment 1, both RTs and accuracy showed an interaction between Probe Referent Order and True Probe. These findings suggest that listeners also maintain spatial information about referents when activation is triggered by a referent property rather than a repeated name. **Conclusions.** Listeners clearly retain incidental spatial information about referents, despite such information not being required for successful task performance. These results demonstrate that spatial representations derived from auditory input play a meaningful role in reference processing.

References:

Emmorey (1996). The Confluence of Space and Language in Signed Languages. In: Bloom, P. (Ed.). *Language and space.* Cambridge, Mass: MIT Press. Chapter 5. (pp. 171-209).

R Core Team (2015). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, Retrieved from http://www.R-project.org/

So, W. C., Kita, S., Goldin-Meadow, S. (2009). Using the hands to identify who does what to whom: Gesture and speech go hand-in-hand. *Cognitive Science*, *33*, 115-125.

Staudte, M. & Altmann, G.T.M. (2017). Recalling what was where when seeing nothing there. Psychonomic Bulletin and Review, 24, 400-407.

Figure 1. shows an illustration of the surround system employed in this experiment. The opaque speakers indicate the ones employed during the study

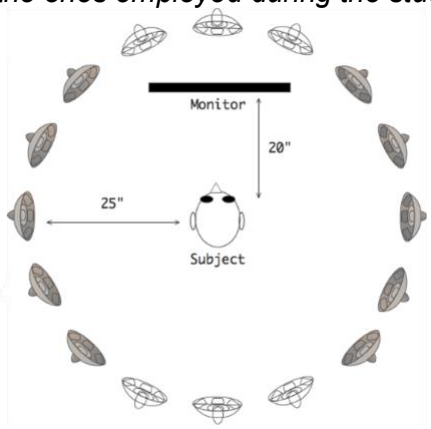

Figure 2. shows an illustration of the procedure employed in Experiment 1 (Top panel) and 2 (Bottom panel).

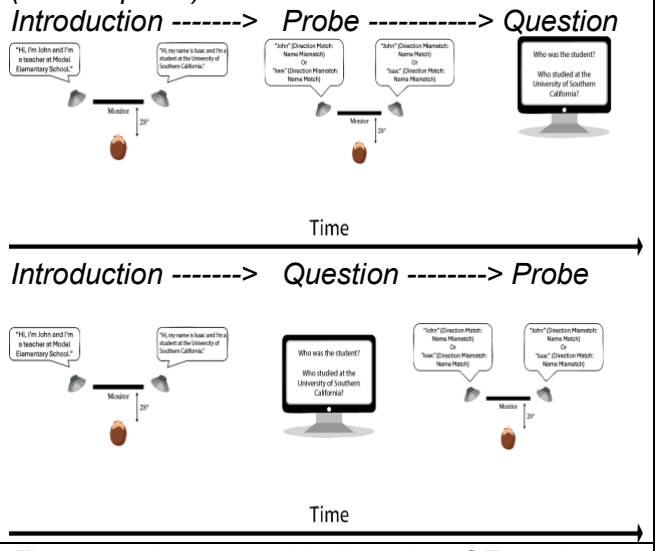

Figure 3. shows graphical results of Experiment 1. The top panel shows Accuracy results. Bottom shows RT.

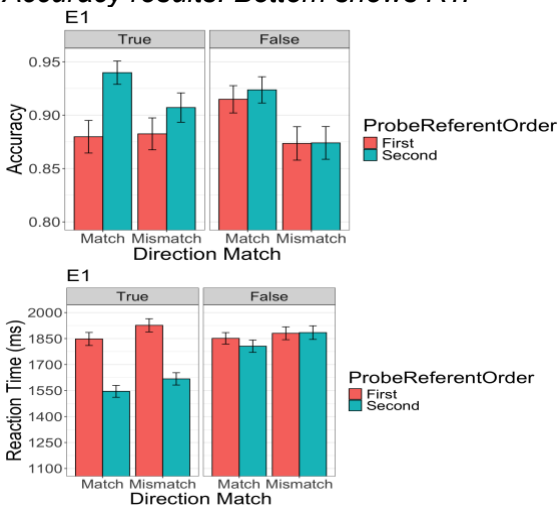
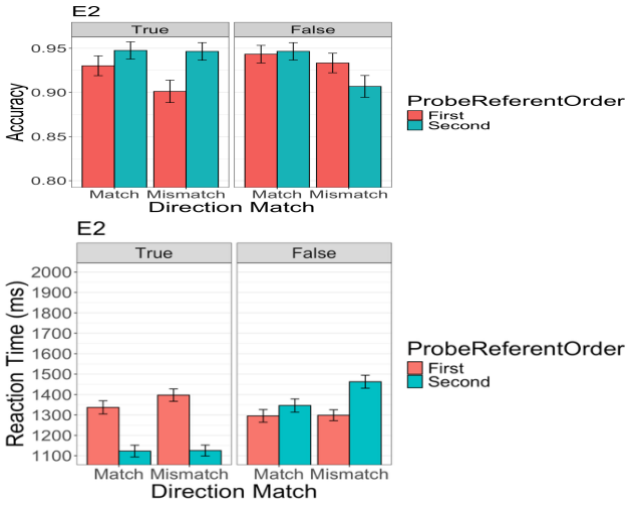
Figure 4. shows graphical results of Experiment 2. The top panel shows Accuracy results. Bottom shows RT.